Module 4 Challenge

Calculus Honors · The Derivative: Definition and Meaning · 16 points

Name: _____ Date: _____

> This is your fullest, most careful work of the module. Show every step in clean vertical layout, justify your reasoning in words where asked, and label your final answers. You will be graded on understanding, quality of reasoning, clarity and precision, and intellectual honesty (say clearly where you are certain and where you are not).

1. Derivative at a point and as a function (6 pts). Let $f(x) = x^2 + 2x$. (a) Use the definition to find $f'(1)$ directly. (b) Use the definition to find the general derivative function $f'(x)$. (c) Confirm that evaluating your part (b) formula at $x = 1$ matches your part (a) result.

2. Tangent line as a rate (5 pts). A weather balloon's altitude (in feet) t minutes after launch is given by $a(t) = 20t^2 + 15t$. (a) Find $a'(3)$ from the definition. (b) Write the equation of the tangent line to $y = a(t)$ at $t = 3$. (c) Interpret the slope of this tangent line as a rate, with correct units, in one sentence.

3. Continuous but not differentiable (5 pts). Let $g(x) = |x + 2|$. (a) Show that g is continuous at $x = -2$. (b) Use one-sided difference quotients to show that g is not differentiable at $x = -2$. (c) Explain, using the general fact that differentiability implies continuity, why this example does not violate that fact. Note clearly how confident you are in each part.